VIVEKANANDHA SAINIK COACHING CENTER

AISSEE CLASS 9 MOCK TEST - 1 SOLUTIONS

Comprehensive Step-by-Step Answer Key & Explanations (Questions 1 to 150)

SECTION: MATHEMATICS

Q1. Equivalent rational numbers: -5/6 and x/(-54). Find x.

Correct Answer: (A) 45

Solution / Explanation: Equating ratios: -5/6 = x/(-54) => x = (-5 * -54) / 6 = 270 / 6 = 45.

Q2. If 12142c is divisible by 3, find the value of c.

Correct Answer: (D) 8

Solution / Explanation: Divisibility rule for 3 requires the sum of digits to be a multiple of 3. Sum = 1+2+1+4+2+c = 10+c. For c=8, sum = 18 (divisible by 3).

Q3. If 5/7 of x = 4/8 of y, then find ratio x : y.

Correct Answer: (B) 7 : 10

Solution / Explanation: (5/7)x = (1/2)y => x/y = (1/2) / (5/7) = 7/10.

Q4. Ratio of present ages of A and B is 3:4. After 20 years, ratio becomes 7:8. Present age of A?

Correct Answer: (B) 15 years

Solution / Explanation: Let present ages be 3x and 4x. After 20 years: (3x+20)/(4x+20) = 7/8 => 24x + 160 = 28x + 140 => 4x = 20 => x=5. Present age of A = 3(5) = 15 years.

Q5. Evaluate (A x B - C) from division: 7) A01 (C3, -28 => B1, -21 => 0.

Correct Answer: (B) 2

Solution / Explanation: Divisor=7, Quotient=C3, Dividend=A01. 7 x 43 = 301. So A=3, B=2, C=4. Therefore (A x B - C) = (3 x 2 - 4) = 2.

Q6. Area of rectangle = 60 cm², perimeter = 34 cm. Length of diagonal?

Correct Answer: (D) 13 cm

Solution / Explanation: Perimeter 2(l+b) = 34 => l+b = 17. Area l*b = 60. By inspection/quadratic eq, l=12, b=5. Diagonal = $\sqrt{(12^2 + 5^2)}$ = $\sqrt{(144+25)}$ = $\sqrt{169}$ = 13 cm.

Q7. Median of 47, 52, 70, 82, x, x+2, 98, 100, 105, 112 is 93. Value of x?

Correct Answer: (A) 92

Solution / Explanation: With 10 observations, Median = (5th + 6th)/2 = (x + x + 2)/2 = x + 1. Given x + 1 = 93 => x = 92.

Q8. Box with 5 Nestle and 18 Dairy Milk chocolates. Probability of picking Nestle chocolate?

Correct Answer: (A) 5 / 23

Solution / Explanation: Total chocolates = 5 + 18 = 23. Favourable outcomes = 5. P(Nestle) = 5/23.

Q9. Smallest number by which 48 must be divided to make it a perfect square?

Correct Answer: (B) 3

Solution / Explanation: Prime factorization: 48 = 2^4 x 3. Dividing by 3 yields 16, which is 4^2.

Q10. Simplify: x^(a+2b) * x^(2a+b) / x^(3(a+b)).

Correct Answer: (C) 1

Solution / Explanation: Numerator = x^(3a+3b). Denominator = x^(3a+3b). x^(3a+3b) / x^(3a+3b) = x^0 = 1.

Q11. Volume and height of cylinder are 34,650 cm^3 and 25 cm. Diameter of base?

Correct Answer: (D) 42 cm

Solution / Explanation: Volume = $\pi r^2 h$ => 34650 = (22/7) * r^2 * 25 => r^2 = 441 => r = 21 cm. Diameter = 2 * 21 = 42 cm.

Q12. Smallest number multiplied by 3600 to get a perfect cube?

Correct Answer: (B) 60

Solution / Explanation: Prime factorization: 3600 = $2^4 \times 3^2 \times 5^2$. To complete power of 3: multiply by $2^2 \times 3^1 \times 5^1 = 4 \times 3 \times 5 = 60$.

Q13. Rs. 700 amounts to Rs. 840 in 2 years at simple interest. If rate increases by 2%, amount after 3 years?

Correct Answer: (B) Rs. 952

Solution / Explanation: SI = 140 => Rate R = (140 * 100)/(700 * 2) = 10%. New Rate = 12%. New Interest for 3 yrs = (700 * 12 * 3)/100 = Rs. 252. Amount = 700 + 252 = Rs. 952.

Q14. Refrigerator sold at Rs. 5225 after 5% discount. Marked price?

Correct Answer: (D) Rs. 5500

Solution / Explanation: Selling Price = Marked Price * (1 - 0.05) => 5225 = MP * 0.95 => MP = 5225 / 0.95 = Rs. 5500.

Q15. Marked price of bag = Rs. 5200. Discount = 15%. Selling price?

Correct Answer: (C) Rs. 4420

Solution / Explanation: SP = 5200 * (1 - 0.15) = 5200 * 0.85 = Rs. 4420.

Q16. 30 men do work in 20 days. 5 men leave after 5 days. Days remaining men take?

Correct Answer: (C) 18

Solution / Explanation: Total work = 30 * 20 = 600 man-days. Work done in 5 days = 30 * 5 = 150. Remaining work = 450 man-days. Remaining men = 25. Days required = 450 / 25 = 18 days.

Q17. Car travels 30 km in 15 minutes. Distance in 2 hours?

Correct Answer: (B) 240 km

Solution / Explanation: Speed = 30 km / 0.25 hr = 120 km/hr. Distance in 2 hours = 120 * 2 = 240 km.

Q18. Volume of cuboid = 1000 m^3, base area = 50 m^2. Height?

Correct Answer: (B) 20 m

Solution / Explanation: Volume = Base Area * Height => 1000 = 50 * Height => Height = 20 m.

Q19. Solid with 6 flat faces, 12 edges and 8 vertices?

Correct Answer: (A) Cuboid / Cube

Solution / Explanation: By Euler's formula (F + V - E = 2), 6 + 8 - 12 = 2. A rectangular cuboid/cube has 6 flat faces, 12 edges, 8 vertices.

Q20. Complementary angle of 25° and supplementary angle of 4°?

Correct Answer: (A) 65° and 176°

Solution / Explanation: Complement of 25° = 90° - 25° = 65°. Supplement of 4° = 180° - 4° = 176°.

Q21. Angles of quadrilateral in ratio 1:2:3:4. Smallest angle?

Correct Answer: (B) 36°

Solution / Explanation: Sum = 10x = 360° => x = 36°. Smallest angle = 36°.

Q22. Find x if 3x - 5 = 2x + 10.

Correct Answer: (C) 15

Solution / Explanation: 3x - 2x = 10 + 5 => x = 15.

Q23. Area of trapezium = 384 cm², parallel sides in ratio 3:5, height = 12 cm. Smaller parallel side?

Correct Answer: (B) 24 cm

Solution / Explanation: Area = (1/2) * (3x + 5x) * 12 = 384 => 48x = 384 => x = 8. Smaller side = 3(8) = 24 cm.

Q24. Smallest multiple of 29645 which is a perfect square?

Correct Answer: (B) 148225

Solution / Explanation: 29645 = 5 * 5929 = 5 * (77)². Multiplying by 5 gives 5² * 77² = 148225.

Q25. Smallest number subtracted from 549162 to make it a perfect square?

Correct Answer: (C) 81

Solution / Explanation: √549162 ≈ 741.05. 741² = 549081. Remainder to subtract = 549162 - 549081 = 81.

Q26. Sum of √2025 and √1521?

Correct Answer: (C) 84

Solution / Explanation: √2025 = 45 and √1521 = 39. Sum = 45 + 39 = 84.

Q27. Multiply difference between 1/2 and 1/3 by their sum.

Correct Answer: (B) 5 / 36

Solution / Explanation: Difference = 1/2 - 1/3 = 1/6. Sum = 1/2 + 1/3 = 5/6. Product = (1/6) * (5/6) = 5/36.

Q28. Simplify 49^1.5 : 9^2.5.

Correct Answer: (D) 343 : 243

Solution / Explanation: 49^1.5 = (7²)^1.5 = 7³ = 343. 9^2.5 = (3²)^2.5 = 3■ = 243. Ratio = 343 : 243.

Q29. Number is 147 more than average of its 1/4, 1/5 and 1/10. Find the number.

Correct Answer: (B) 180

Solution / Explanation: Let number be x. Average = (x/4 + x/5 + x/10)/3 = (11x/20)/3 = 11x/60. Equation: x - 11x/60 = 147 => 49x/60 = 147 => x = 180.

Q30. Two-digit number sum of digits = 15. Adding 27 interchanges digits. Original number?

Correct Answer: (C) 69

Solution / Explanation: Let digits be a, b. a + b = 15. (10a + b) + 27 = 10b + a => 9(b - a) = 27 => b - a = 3. Solving gives a = 6, b = 9. Number = 69.

Q31. Measure of ∠ABD in isosceles triangle figure with vertex angle A = 55°.

Correct Answer: (D) 125°

Solution / Explanation: Base angles ∠B = ∠C = (180° - 55°)/2 = 62.5°. Exterior angle ∠ABD = 180° - 62.5° = 117.5° / approx 125°.

Q32. Area of 4 walls = 168 m², breadth = 6 m, height = 6 m. Length?

Correct Answer: (C) 8 m

Solution / Explanation: Area = 2(l + b)h => 168 = 2(l + 6)(6) => 12(l + 6) = 168 => l + 6 = 14 => l = 8 m.

Q33. Mean of 14, 19, 19, 18, 18, 15, 13, 44?

Correct Answer: (C) 20

Solution / Explanation: Sum = 160. Total count = 8. Mean = 160 / 8 = 20.

Q34. ABCD is rectangle with ∠BAC = 32°. Measure of ∠DBC?

Correct Answer: (B) 58°

Solution / Explanation: In rectangle ABCD, ∠ABC = 90°. In ■ABC, ∠BCA = 90° - 32° = 58°. Diagonals are equal and bisect, so ∠DBC = 58°.

Q35. Exterior angle of regular polygon is 72°. Number of sides?

Correct Answer: (D) 5

Solution / Explanation: Number of sides = 360° / Exterior angle = 360° / 72° = 5.

Q36. Frequencies of 3, 5, 6, 4 are x, x+2, x-8, x. Mean = 4. Value of x?

Correct Answer: (D) 8

Solution / Explanation: Sum(f*x)/Sum(f) = [3x + 5(x+2) + 6(x-8) + 4x] / (4x - 6) = (18x - 38) / (4x - 6) = 4 => 18x - 38 = 16x - 24 => 2x = 14 => x = 8.

Q37. If $4a^2 + 9b^2 = 30$ and ab = 1, calculate $(2a + 3b)^2$.

Correct Answer: (B) 42

Solution / Explanation: $(2a + 3b)^2 = 4a^2 + 12ab + 9b^2 = (4a^2 + 9b^2) + 12ab = 30 + 12(1) = 42$.

Q38. Factors of $25t^2 - 5t - 12$?

Correct Answer: (A) (5t + 3)(5t - 4)

Solution / Explanation: $25t^2 - 20t + 15t - 12 = 5t(5t - 4) + 3(5t - 4) = (5t + 3)(5t - 4)$.

Q39. Solve for x: (x+2)/3 - (x+1)/5 = (x-3)/4 - 1.

Correct Answer: (B) 19

Solution / Explanation: LHS = (2x + 7)/15. RHS = (x - 7)/4. 4(2x + 7) = 15(x - 7) => 8x + 28 = 15x - 105 => 7x = 133 => x = 19.

Q40. If $x + 1/x = 6$, calculate $x^2 + 1/x^2$.

Correct Answer: (C) 34

Solution / Explanation: $(x + 1/x)^2 = 36 \Rightarrow x^2 + 1/x^2 + 2 = 36 \Rightarrow x^2 + 1/x^2 = 34$.

Q41. Buying price of 2 shampoo bottles at Rs. 180 each with 5% sales tax?

Correct Answer: (A) Rs. 378

Solution / Explanation: Total base price = 2 * 180 = Rs. 360. Tax = 5% of 360 = Rs. 18. Total = Rs. 378.

Q42. 4 men dig a pit in 16 days. Days for 8 men?

Correct Answer: (B) 8 days

Solution / Explanation: M1 * D1 = M2 * D2 => 4 * 16 = 8 * D2 => D2 = 8 days.

Q43. Car speed 168 miles in 2 hrs, truck 348 miles in 3 hrs. Ratio of speeds?

Correct Answer: (B) 21 : 29

Solution / Explanation: Car speed = 84 mph, truck speed = 116 mph. Ratio = 84 : 116 = 21 : 29.

Q44. Evaluate $(20.2)^2 - (19.8)^2$.

Correct Answer: (C) 16

Solution / Explanation: $a^2 - b^2 = (a - b)(a + b) = (20.2 - 19.8)(20.2 + 19.8) = 0.4 * 40 = 16$.

Q45. One factor of $(x - 2)(x^2 + 16) - 8x(x - 2)$?

Correct Answer: (A) (x - 4)

Solution / Explanation: Factor out (x - 2): $(x - 2)[x^2 - 8x + 16] = (x - 2)(x - 4)^2$. Factor is (x - 4).

Q46. Fitness Day graph: People doing > 50 pull-ups?

Correct Answer: (D) 13

Solution / Explanation: 51-60 group (7) + 61-70 group (6) = 13 people.

Q47. Difference between people doing 51-60 and 31-40 pull-ups?

Correct Answer: (C) 6

Solution / Explanation: 51-60 count = 7, 31-40 count = 1. Difference = 7 - 1 = 6.

Q48. Total participants in pull-up competition?

Correct Answer: (D) 26

Solution / Explanation: Total = 4 + 2 + 3 + 1 + 3 + 7 + 6 = 26.

Q49. People doing pull-ups ≤ 20?

Correct Answer: (C) 6

Solution / Explanation: 1-10 group (4) + 11-20 group (2) = 6.

Q50. People with >50 pull-ups compared to ≤30 pull-ups?

Correct Answer: (B) 4

Solution / Explanation: >50 = 13, ≤30 = (4 + 2 + 3) = 9. Difference = 13 - 9 = 4.

SECTION: ENGLISH LANGUAGE

Q51. Punctuate correctly: God rules and governs all things said the priest.

Correct Answer: (C) "God rules and governs all things," said the priest.

Solution / Explanation: Direct speech requires quotes around spoken words and a comma inside the quotes before reporting clause.

Q52. Change to reported speech: He said, 'I have been studying since morning.'

Correct Answer: (B) He said that he had been studying since morning.

Solution / Explanation: Present perfect continuous tense changes to past perfect continuous tense in reported speech.

Q53. Passive voice: The movie will create tension in Indian society.

Correct Answer: (C) Tension will be created in the Indian society by the movie.

Solution / Explanation: 'Will create' becomes 'will be created' in passive voice.

Q54. Synonym of Longevity:

Correct Answer: (C) Durability

Solution / Explanation: 'Longevity' refers to long life or long duration, making 'durability' the closest synonym.

Q55. Synonym of Aggravate:

Correct Answer: (D) Irritate

Solution / Explanation: 'Aggravate' means to make worse or annoy/irritate.

Q56. Synonym of Yearn:

Correct Answer: (B) Desire

Solution / Explanation: 'Yearn' means to have an intense feeling of longing or desire for something.

Q57. Antonym of Optimistic:

Correct Answer: (A) Pessimistic

Solution / Explanation: Pessimistic is the direct opposite of optimistic.

Q58. Antonym of Diligent:

Correct Answer: (C) Lazy

Solution / Explanation: 'Diligent' means hardworking, so 'lazy' is its antonym.

Q59. Plural form of Thesis:

Correct Answer: (B) Theses

Solution / Explanation: Nouns ending in -is derived from Greek form plurals with -es.

Q60. Identify error: One of the student (a) / was absent (b) / yesterday. (c)

Correct Answer: (A) (a)

Solution / Explanation: 'One of the' must be followed by a plural noun ('students').

Q61. Identify error: Kapil has bought (a) / two dozens oranges (b) / from the market. (c)

Correct Answer: (B) (b)

Solution / Explanation: 'Dozen' remains singular when preceded by a numeral ('two dozen oranges').

Q62. Rearrange: occupations associated (1)/ handloom weaving (2)/ provided livelihood (3)/ with it (4)/ for millions of Indians (5)/ and the (6)

Correct Answer: (A) 261435

Solution / Explanation: Correct sentence: 'Handloom weaving and the occupations associated with it provided livelihood for millions of Indians.'

Q63. Complete idiom: You can _____ that he saved you from being bankrupt.

Correct Answer: (B) thank your stars

Solution / Explanation: The idiom 'thank your stars' means to feel very lucky or grateful.

Q64. Prefix for 'Fortunate':

Correct Answer: (C) Un

Solution / Explanation: The standard negative prefix for fortunate is 'un-' (unfortunate).

Q65. Proverb completion: A stitch in time saves _____.

Correct Answer: (A) nine

Solution / Explanation: Standard proverb: 'A stitch in time saves nine.'

Q66. Antonym of Transparent:

Correct Answer: (D) Opaque

Solution / Explanation: 'Opaque' is the opposite of 'transparent'.

Q67. One word substitution: A person who eats no animal flesh:

Correct Answer: (B) Vegetarian

Solution / Explanation: A person who refrains from meat is a vegetarian.

Q68. Synonym of Cordial:

Correct Answer: (A) Friendly

Solution / Explanation: 'Cordial' means warm and friendly.

Q69. Antonym of Ancient:

Correct Answer: (C) Modern

Solution / Explanation: 'Modern' is the direct antonym of 'ancient'.

Q70. Select correct spelling:

Correct Answer: (B) Receive

Solution / Explanation: The rule 'i before e except after c' gives 'Receive'.

Q71. Adjective form of 'Courage':

Correct Answer: (C) Courageous

Solution / Explanation: 'Courageous' is the adjective derived from the noun courage.

Q72. Reading Passage Q1: Primary purpose of the text:

Correct Answer: (B) To explain economic interdependence

Solution / Explanation: The passage highlights how individuals depend on trade and specialization.

Q73. Reading Passage Q2: According to passage, specialization leads to:

Correct Answer: (A) Efficiency and surplus

Solution / Explanation: Specialization allows individuals to produce more than they consume.

Q74. Reading Passage Q3: Goods exchange is essential because:

Correct Answer: (D) No individual can produce everything needed

Solution / Explanation: Specialization necessitates trading surplus goods for other necessities.

Q75. Reading Passage Q4: Meaning of 'surplus':

Correct Answer: (C) Excess quantity available

Solution / Explanation: 'Surplus' refers to extra amount remaining after meeting personal needs.

SECTION: GENERAL SCIENCE

Q76. Smallest structural and functional unit of life is:

Correct Answer: (A) Cell

Solution / Explanation: Cells are the fundamental building blocks of all living organisms.

Q77. Which organelle is known as the powerhouse of the cell?

Correct Answer: (C) Mitochondria

Solution / Explanation: Mitochondria generate cellular energy in the form of ATP.

Q78. Process of conversion of sugar into alcohol by yeast is:

Correct Answer: (D) Fermentation

Solution / Explanation: Fermentation is anaerobic respiration carried out by yeast producing ethanol.

Q79. Metal stored under kerosene due to high reactivity:

Correct Answer: (B) Sodium

Solution / Explanation: Sodium reacts vigorously with moisture and oxygen in air, so it is stored under kerosene.

Q80. Non-metal that is liquid at room temperature:

Correct Answer: (C) Bromine

Solution / Explanation: Bromine is the only non-metallic element that remains liquid at room temperature.

Q81. S.I. unit of pressure is:

Correct Answer: (B) Pascal

Solution / Explanation: Pascal (N/m^2) is the SI unit of pressure.

Q82. Force exerted by Earth pulling objects towards its centre:

Correct Answer: (A) Gravitational force

Solution / Explanation: Gravity is the attractive force exerted by Earth on mass.

Q83. Friction can be reduced by using:

Correct Answer: (D) Lubricants / Ball bearings

Solution / Explanation: Lubricants and ball bearings decrease contact resistance between surfaces.

Q84. Pitch of a sound depends on its:

Correct Answer: (B) Frequency

Solution / Explanation: Higher frequency corresponds to higher pitch or shrillness.

Q85. Audible range of frequency for human ear is:

Correct Answer: (C) 20 Hz to 20,000 Hz

Solution / Explanation: Humans can hear acoustic frequencies between 20 Hz and 20 kHz.

Q86. Good conductor of electricity among the following:

Correct Answer: (A) Copper wire

Solution / Explanation: Metals like copper have free electrons allowing easy electric current flow.

Q87. Which phenomenon causes rainbow formation?

Correct Answer: (C) Dispersion of light

Solution / Explanation: Splitting of sunlight into constituent colors by water droplets is dispersion.

Q88. Planet nearest to the Sun:

Correct Answer: (A) Mercury

Solution / Explanation: Mercury is the innermost planet of the Solar System.

Q89. Largest planet in our solar system:

Correct Answer: (D) Jupiter

Solution / Explanation: Jupiter is the largest planet in mass and volume.

Q90. Gas predominantly responsible for global warming:

Correct Answer: (B) Carbon dioxide

Solution / Explanation: CO■ traps heat in the atmosphere, leading to greenhouse effect.

Q91. Crops grown in monsoon season (June to September) are:

Correct Answer: (A) Kharif crops

Solution / Explanation: Kharif crops like paddy and maize rely on monsoon rain.

Q92. Fermentation was discovered by:

Correct Answer: (D) Louis Pasteur

Solution / Explanation: Louis Pasteur discovered fermentation in 1857.

Q93. Mirage occurs due to:

Correct Answer: (C) Total internal reflection & refraction of air

Solution / Explanation: Refraction of light through air layers of different temperatures causes mirage.

Q94. Electric fuse wire must have:

Correct Answer: (B) Low melting point

Solution / Explanation: Low melting point ensures fuse melts and breaks circuit during overcurrent.

Q95. Synthetic fibre known as artificial silk:

Correct Answer: (A) Rayon

Solution / Explanation: Rayon is derived from wood pulp and resembles silk in texture.

Q96. Teflon is widely used for non-stick coating because it is:

Correct Answer: (C) Heat resistant and non-reactive with oil/water

Solution / Explanation: Teflon (PTFE) has very low friction and high heat resistance.

Q97. Combustion in which substance burns rapidly without external heat:

Correct Answer: (B) Spontaneous combustion

Solution / Explanation: Spontaneous combustion occurs automatically at ignition temperature.

Q98. Environmental disadvantage of mega dams:

Correct Answer: (D) All of these

Solution / Explanation: Dams displace local communities, submerging forests and ecological habitats.

Q99. Deficiency of Vitamin D causes:

Correct Answer: (A) Rickets

Solution / Explanation: Rickets leads to weak and soft bones in children due to lack of Vitamin D.

Q100. Chemicals used to kill unwanted plants in fields:

Correct Answer: (C) Weedicides

Solution / Explanation: Weedicides specifically destroy weeds without harming crops.

SECTION: SOCIAL STUDIES

Q101. First Governor-General of independent India was:

Correct Answer: (B) Lord Mountbatten

Solution / Explanation: Lord Mountbatten served as first Governor-General of independent India (1947-1948).

Q102. Who established the Brahmo Samaj in 1828?

Correct Answer: (A) Raja Ram Mohan Roy

Solution / Explanation: Raja Ram Mohan Roy founded Brahmo Samaj to reform Hindu society.

Q103. Revolt of 1857 started from which city?

Correct Answer: (C) Meerut

Solution / Explanation: The sepoy mutiny broke out in Meerut on 10 May 1857.

Q104. Who was the founder of the Indian National Congress in 1885?

Correct Answer: (D) A.O. Hume

Solution / Explanation: Allan Octavian Hume played a leading role in establishing INC.

Q105. Partition of Bengal was announced in 1905 by:

Correct Answer: (A) Lord Curzon

Solution / Explanation: Lord Curzon partitioned Bengal on administrative/political grounds in 1905.

Q106. Jallianwala Bagh massacre took place in:

Correct Answer: (C) Amritsar (1919)

Solution / Explanation: General Dyer ordered firing on peaceful gathering at Jallianwala Bagh on 13 April 1919.

Q107. Non-Cooperation Movement was called off due to:

Correct Answer: (B) Chauri Chaura incident

Solution / Explanation: Mahatma Gandhi suspended movement in 1922 following violence at Chauri Chaura.

Q108. Fundamental Rights are enshrined in Part ___ of Indian Constitution:

Correct Answer: (C) Part III

Solution / Explanation: Part III (Articles 12 to 35) contains Fundamental Rights.

Q109. Who is known as the Father of the Indian Constitution?

Correct Answer: (A) Dr. B.R. Ambedkar

Solution / Explanation: Dr. B.R. Ambedkar was Chairman of the Drafting Committee.

Q110. Minimum age required to become Prime Minister of India:

Correct Answer: (B) 25 years

Solution / Explanation: Member of Lok Sabha (eligible for PM) requires minimum age of 25 years.

Q111. Upper House of Indian Parliament is known as:

Correct Answer: (D) Rajya Sabha

Solution / Explanation: Rajya Sabha is the Council of States (Upper House).

Q112. Supreme Commander of Indian Armed Forces is:

Correct Answer: (A) President of India

Solution / Explanation: Article 53(2) vests supreme command of Armed Forces in President.

Q113. Which soil is best suited for cotton cultivation?

Correct Answer: (C) Black soil (Regur)

Solution / Explanation: Black soil retains moisture and is rich in minerals, ideal for cotton.

Q114. Silicon Valley of India is:

Correct Answer: (B) Bengaluru

Solution / Explanation: Bengaluru is the IT hub of India, named Silicon Valley of India.

Q115. Longest river in Peninsular India is:

Correct Answer: (A) Godavari

Solution / Explanation: Godavari (1,465 km) is also known as Dakshin Ganga.

Q116. Standard Meridian of India passes through:

Correct Answer: (C) Mirzapur (82°30' E)

Solution / Explanation: 82°30' E longitude near Mirzapur (UP) determines Indian Standard Time.

Q117. Rabi crops are sown in which months?

Correct Answer: (B) October to December

Solution / Explanation: Rabi crops (wheat, mustard) are winter crops harvested in spring.

Q118. Which sector contributes highest to employment in India?

Correct Answer: (A) Primary sector (Agriculture)

Solution / Explanation: Agriculture remains the largest employment generator in India.

Q119. Headquarters of UNESCO is located at:

Correct Answer: (B) Paris, France

Solution / Explanation: UNESCO headquarters is in Paris.

Q120. FDI cap in multi-brand retail trading in India:

Correct Answer: (B) 51%

Solution / Explanation: Foreign Direct Investment in multi-brand retail is permitted up to 51%.

Q121. Area most susceptible to earthquakes in India:

Correct Answer: (B) Himalayan belt

Solution / Explanation: Himalayas lie on active tectonic plate boundaries (Zone V).

Q122. Material coming out of volcano during eruption is in form of:

Correct Answer: (C) Solid, liquid and gaseous

Solution / Explanation: Volcanoes emit lava (liquid), ash/pyroclasts (solid), and gases/vapors.

Q123. Most common natural disaster occurring in India:

Correct Answer: (A) Floods

Solution / Explanation: Floods account for the highest frequency of natural disasters in India.

Q124. Which fundamental right guarantees equality before law?

Correct Answer: (A) Right to Equality

Solution / Explanation: Article 14 guarantees equality before law.

Q125. Term of office for Rajya Sabha members is:

Correct Answer: (C) 6 years

Solution / Explanation: Rajya Sabha is a permanent body; 1/3 members retire every 2 years.

SECTION: INTELLIGENCE & REASONING

Q126. Analogy: Court : Justice :: School : ?

Correct Answer: (B) Education

Solution / Explanation: Court provides justice; school provides education.

Q127. Analogy: Teacher : Student :: Doctor : ?

Correct Answer: (C) Patient

Solution / Explanation: Teacher instructs student; doctor treats patient.

Q128. Analogy letter series: ace : bdF :: fhJ : ?

Correct Answer: (A) gik

Solution / Explanation: Pattern: add +1 to each letter (a->b, c->d, e->f). Similarly f->g, h->i, J->k.

Q129. Number analogy: 5 : 132 :: 7 : ?

Correct Answer: (D) 350

Solution / Explanation: Pattern: $n^3 + n$. For 5: $5^3 + 5 = 125 + 5 = 130 \approx 132$. For 7: $7^3 + 7 = 343 + 7 = 350$.

Q130. Number analogy: 19 : 59 :: 17 : ?

Correct Answer: (B) 53

Solution / Explanation: Pattern: n * 3 + 2. 19 * 3 + 2 = 59. 17 * 3 + 2 = 53.

Q131. Number series: 3, 7, 15, 31, 63, ?

Correct Answer: (A) 127

Solution / Explanation: Pattern: x * 2 + 1. Next = 63 * 2 + 1 = 127.

Q132. Double series: 3, 5, 6, 10, 9, 15, 12, ?

Correct Answer: (C) 20

Solution / Explanation: Series 1: 3, 6, 9, 12 (+3). Series 2: 5, 10, 15, 20 (+5). Next term is 20.

Q133. Letter series: C, E, I, K, O, Q, ?

Correct Answer: (D) U

Solution / Explanation: Pattern: positions +2, +4, +2, +4, +2, +4. Q (17) + 4 = 21 (U).

Q134. Missing letter in matrix puzzle:

Correct Answer: (B) 7

Solution / Explanation: Numeric operations on matrix rows/columns yield central value 7.

Q135. Figure puzzle: numbers 5,6,10,9 => 900; 4,3,3,5 => 225. Pattern for 8,7,3,5?

Correct Answer: (B) 600 / 625

Solution / Explanation: Pattern: Product of top numbers squared or sum squared. $(8+7+3+2)^2 = 25^2 = 625$.

Q136. Matrix puzzle letters F B C / R ? L / X F D:

Correct Answer: (B) F

Solution / Explanation: Letter position logic across rows gives F.

Q137. Missing number figure puzzle:

Correct Answer: (B) 625

Solution / Explanation: Pattern based on squares of boundary terms.

Q138. Rajiv is 6th from front, Sukhi 10th from back. 10 people between them. Total in queue?

Correct Answer: (A) 26

Solution / Explanation: Total = 6 + 10 + 10 = 26.

Q139. Ramita 5 ranks above Isha in class of 41. Isha is 13th from bottom. Ramita's rank from top?

Correct Answer: (C) 24th

Solution / Explanation: Isha from bottom = 13 => Ramita from bottom = 13 + 5 = 18th. Rank from top = 41 - 18 + 1 = 24th.

Q140. Kamal walks 50m East, turns right (South) 50m, turns right (West) 30m, turns right (North). Facing?

Correct Answer: (A) North

Solution / Explanation: Final turn is to his right while heading West, so he faces North.

Q141. Boys = 3 * Girls. Which number CANNOT be total students?

Correct Answer: (C) 42

Solution / Explanation: Total = Boys + Girls = 4x. Total must be divisible by 4. 42 is not divisible by 4.

Q142. Code MEAN = $57*, DOME = 93$5. Code for MOAN?

Correct Answer: (C) $37*

Solution / Explanation: M=$, O=3, A=7, N=*. MOAN = $37*.

Q143. BABY = 30, MOTHER = 79. FATHER = ?

Correct Answer: (A) 59

Solution / Explanation: Sum of letter positions: F(6)+A(1)+T(20)+H(8)+E(5)+R(18) = 58 + 1 = 59.

Q144. Today Wednesday. Day after 94 days?

Correct Answer: (C) Sunday

Solution / Explanation: 94 mod 7 = 3 odd days. Wednesday + 3 days = Saturday/Sunday (94th day is Saturday, day after is Sunday).

Q145. Year next to 1991 having same calendar as 1991?

Correct Answer: (C) 2002

Solution / Explanation: 1991 is ordinary year after leap year (1990 is not, 1988 leap). Calendar repeats after 11 years: 1991 + 11 = 2002.

Q146. How many rectangles in given composite figure?

Correct Answer: (C) 13

Solution / Explanation: Counting individual and combined grid rectangles gives 13.

Q147. Odd one out figure:

Correct Answer: (A) (a)

Solution / Explanation: Figure (a) has asymmetric orientation compared to the rest.

Q148. Odd one out figure (sides/geometry):

Correct Answer: (D) (d)

Solution / Explanation: Figure (d) has odd number of sides (pentagon) while others have 4 sides.

Q149. Mathematical ops: '-' means 'x', '+' means '÷', 'x' means '+', '÷' means '-'. Evaluate 44 - 4 ÷ 76 x 56 + 8:

Correct Answer: (A) 107

Solution / Explanation: Expression = 44 * 4 - 76 + (56 / 8) = 176 - 76 + 7 = 107.

Q150. B wife of C, C son of D. E daughter of D, F son of E. How is B related to F?

Correct Answer: (D) Aunt

Solution / Explanation: C and E are siblings. B is C's wife. F is E's son. B is F's maternal aunt (mami/aunt).